

README2 - Project Knowledge Base

This document summarizes the Blood Bank MERN project in detail so a chatbot can answer user questions about the site, its pages, flows, forms, and endpoints.

Project Identity

App names in UI: Blood Axis, Blood Bank App, and Blood Bank Nepal.

Purpose: Manage blood inventory, donor activity, hospital and organization interactions, handle blood requests and payments, and provide blood demand forecasting.

Tech and Frontend Libraries

React app created with Create React App.

UI frameworks: Bootstrap 5.3.0 alpha 3, MUI.

Icons: Font Awesome 6.4.0, MUI icons, react-icons.

Charts: Chart.js and react-chartjs-2.

Date formatting: moment.

Routing: react-router-dom.

Data fetching: @tanstack/react-query.

Toast notifications: react-toastify.

Authentication and Access Control

Protected routes require a token in localStorage.

Public routes redirect authenticated users to /home.

Role-based redirects:

admin -> /admin

donar -> /donor-dashboard

Others -> inventory dashboard.

Payment Flow Summary

eSewa Flow:

Hospital submits request supply form.

App calls eSewa initiate endpoint.

eSewa redirects to /payment-success.

Payment is verified and request is created with paymentStatus: paid.

Khalti Flow:

User submits blood payment request form.

App calls Khalti initiate endpoint.

Khalti redirects to success page with pidx and purchase_order_id.

Payment is verified and user is redirected to /donation.

Forecasting Module

Predicts 7-day blood demand by blood group for each organization.

Uses SARIMA with fallback models.

Provides forecast endpoints, history, stats, and admin scheduler control.

API Endpoints Summary

Auth:

GET /auth/current-user

PUT /auth/update-profile

Inventory:

GET /inventory/get-inventory

GET /inventory/get-recent-inventory

GET /inventory/get-donars

GET /inventory/get-hospitals

Blood Requests:

POST /request/create-request

GET /request/get-requests

PUT /request/update-status/:id

PUT /request/approve/:id

PUT /request/reject/:id

Admin:

GET /admin/donar-list

GET /admin/hospital-list

GET /admin/org-list

DELETE /admin/delete-donar/:id

Payments:

POST /api/v1/esewa/initiate

POST /api/v1/esewa/verify

POST /api/v1/khalti/initiate

POST /api/v1/khalti/verify

This file contains structured knowledge of the Blood Bank Nepal MERN project including routes, dashboards, donor flow, hospital flow, admin flow, payments, forecasting, and directory data.